

JEE Mains 2019 Chapter wise Question Bank

Semiconductors - Questions

Q1

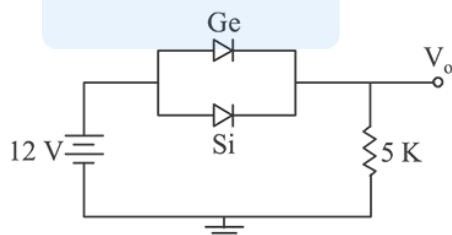
Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n-type semiconductor, the density of electrons is 10^{19} m^{-3} and their mobility is $1.6 \text{ m}^2/(\text{V.s})$ then the resistivity of the semiconductor (since it is an n-type semiconductor contribution of holes is ignored) is close to:

- (1) $2 \Omega\text{m}$ (2) $4 \Omega\text{m}$
 (3) $0.4 \Omega\text{m}$ (4) $0.2 \Omega\text{m}$

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Q2

Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of V_o changes by : (assume that the Ge diode has large breakdown voltage)

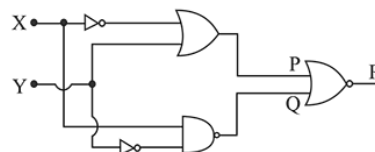


- (1) 0.8 V (2) 0.6 V
 (3) 0.2 V (4) 0.4 V

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Q3

To get output '1' at R, for the given logic gate circuit the input values must be:

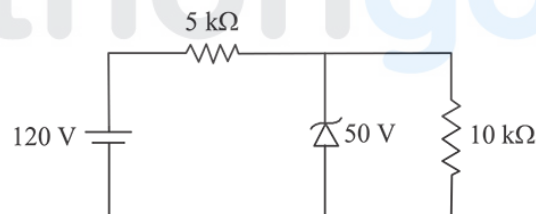


- (1) $X = 0, Y = 1$ (2) $X = 1, Y = 1$
 (3) $X = 1, Y = 0$ (4) $X = 0, Y = 0$

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Q4

For the circuit shown below, the current through the Zener diode is:

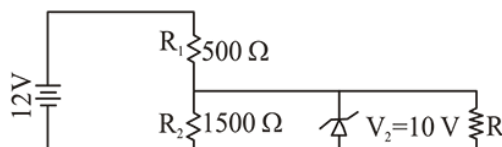


- (1) 9 mA (2) 5 mA
 (3) Zero (4) 14 mA

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Q5

In the given circuit the current through Zener Diode is close to :



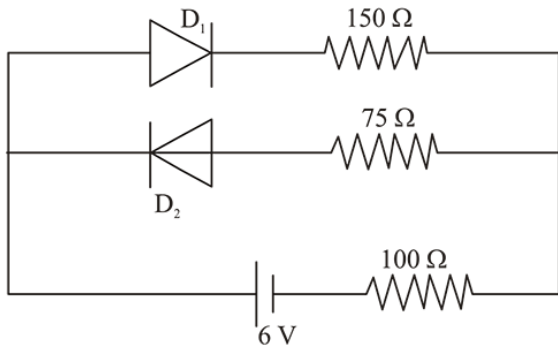
- (1) 0.0 mA (2) 6.7 mA
 (3) 4.0 mA (4) 6.0 mA

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Semiconductors

Q6

The circuit shown below contains two ideal diodes, each with a forward resistance of $50\ \Omega$. If the battery voltage is 6 V , the current through the $100\ \Omega$ resistance (in Amperes) is :

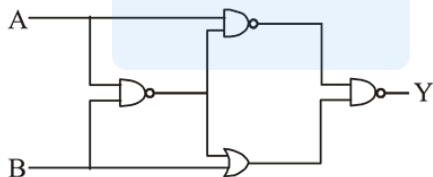


- (1) 0.036 (2) 0.020
(3) 0.027 (4) 0.030

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Q7

The output of the given logic circuit is:

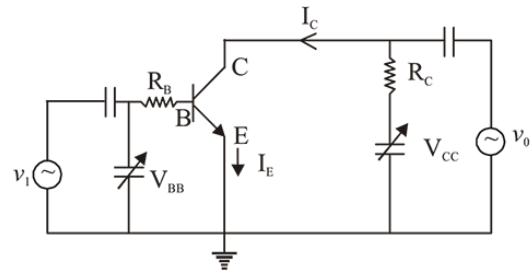


- (1) $A\bar{B} + \bar{A}B$ (2) $AB + \bar{A}\bar{B}$
(3) $A\bar{B}$ (4) $\bar{A}B$

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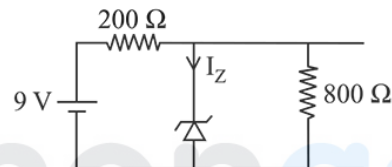
In the figure, given that V_{BE} supply can vary from 0 to 5.0 V , $V_{CE} = 5\text{ V}$, $\beta_{dc} = 200$, $R_B = 100\text{ k}\Omega$, $R_C = 1\text{ k}\Omega$ and $V_{BE} = 1.0\text{ V}$. The minimum base current and the input voltage at which the transistor will go to saturation, will be, respectively :

- (1) $25\ \mu\text{A}$ and 3.5 V (2) $20\ \mu\text{A}$ and 3.5 V
(3) $25\ \mu\text{A}$ and 2.8 V (4) $20\ \mu\text{A}$ and 2.8 V

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Q9

The reverse breakdown voltage of a Zener diode is 5.6 V in the given circuit.



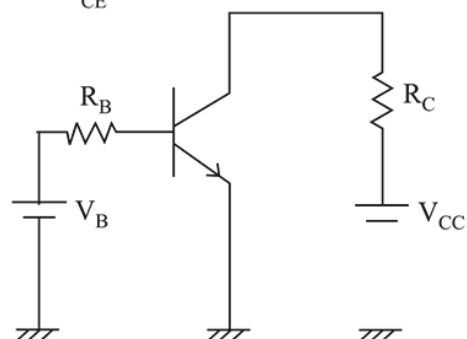
The current I_z through the Zener is :

- (1) 10 mA (2) 17 mA
(3) 15 mA (4) 7 mA

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Q10

A common emitter amplifier circuit, built using an npn transistor, is shown in the figure. Its dc current gain is 250, $R_C = 1\text{ k}\Omega$ and $V_{CC} = 10\text{ V}$. What is the minimum base current for V_{CE} to reach saturation ?



- (1) $40\ \mu\text{A}$ (2) $100\ \mu\text{A}$ (3) $7\ \mu\text{A}$ (4) $10\ \mu\text{A}$

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Semiconductors

Q11

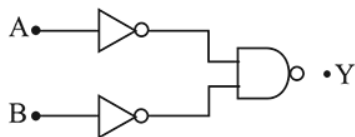
An NPN transistor is used in common emitter configuration as an amplifier with $1\text{ k}\Omega$ load resistance. Signal voltage of 10 mV is applied across the base-emitter. This produces a 3 mA change in the collector current and $15\text{ }\mu\text{A}$ change in the base current of the amplifier. The input resistance and voltage gain are:

- (1) $0.33\text{ k}\Omega$, 1.5 (2) $0.67\text{ k}\Omega$, 300
(3) $0.67\text{ k}\Omega$, 200 (4) $0.33\text{ k}\Omega$, 300

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Q11

The logic gate equivalent to the given logic circuit is:



- (1) NAND (2) OR (3) NOR (4) AND

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Q12

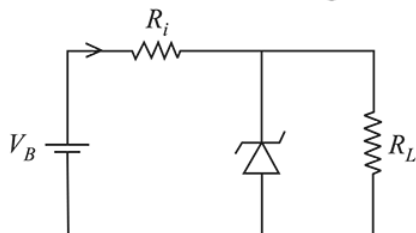
An npn transistor operates as a common emitter amplifier, with a power gain of 60 dB . The input circuit resistance is 100Ω and the output load resistance is $10\text{ k}\Omega$. The common emitter current gain β is :

- (1) 10^2 (2) 60 (3) 6×10^2 (4) 10^4

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Q13

The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is $R_L = 4\text{ k}\Omega$. The series resistance of the circuit is $R_i = 1\text{ k}\Omega$. If the battery voltage V_B varies from 8 V to 16 V , what are the minimum and maximum values of the current through Zener diode ?



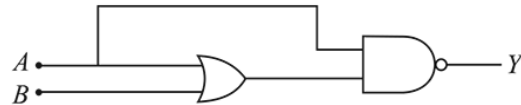
- (1) 0.5 mA ; 6 mA (2) 1 mA ; 8.5 mA
(3) 0.5 mA ; 8.5 mA (4) 1.5 mA ; 8.5 mA

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Q14

The truth table for the circuit given in the fig. is :



(1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

(2)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(3)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

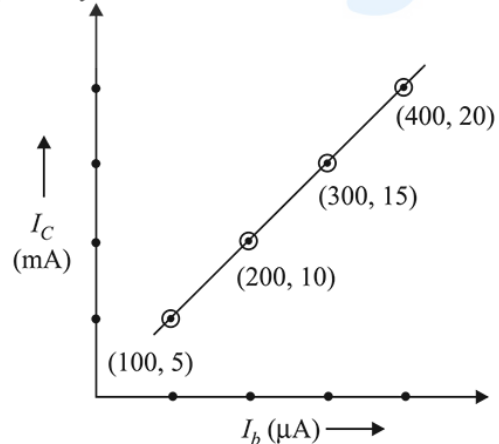
(4)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

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Q14

The transfer characteristic curve of a transistor, having input and output resistance 100Ω and $100\text{ k}\Omega$ respectively, is shown in the figure. The Voltage and Power gain, are respectively :

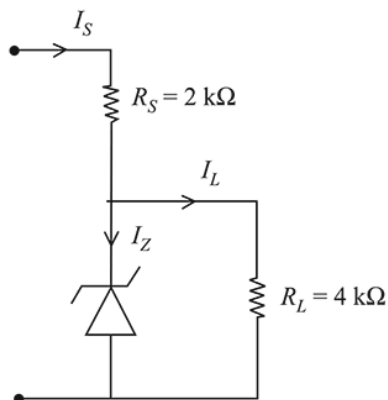


- (1) 2.5×10^4 , 2.5×10^6 (2) 5×10^4 , 5×10^6
(3) 5×10^4 , 5×10^5 (4) 5×10^4 , 2.5×10^6

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Q15

Figure shows a DC voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current ?



- (1) 2.5 mA (2) 1.5 mA (3) 7.5 mA (4) 3.5 mA

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Semiconductors - Answers

Q1

(3) As we know, current density,

$$j = \sigma E = nev_d$$

$$\sigma = ne \frac{v_d}{E} = ne\mu$$

$$\frac{1}{\sigma} = \rho = \frac{1}{n_e e \mu_e} = \text{Resistivity}$$

$$= \frac{1}{10^{19} \times 1.6 \times 10^{19} \times 19 \times 1.6}$$

$$\text{or } \rho = 0.4 \Omega \text{m}$$

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Q2

(4) Initially Ge and Si are both forward biased so current will effectively pass through Ge diode $\therefore V_o = 12 - 0.3 = 11.7 \text{ V}$

And if "Ge" is reversed then current will flow through "Si" diode

$$\therefore V_o = 12 - 0.7 = 11.3 \text{ V}$$

Clearly, V_o changes by $11.7 - 11.3 = 0.4 \text{ V}$

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Q3

(3) From the given logic circuit,

$$p = \bar{x} + y$$

$$Q = \overline{\bar{y} \cdot x} = y + \bar{x}$$

$$\text{Output, } R = \overline{p + Q}$$

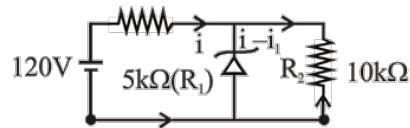
To make output 1

 $P + Q$ must be '0'So, $x = 1, y = 0$

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Q4

(1) The voltage across zener diode is constant



$$i_{(R_2)} = \frac{V}{R} = \frac{50}{10 \times 10^3} = 5 \times 10^{-3} \text{ A}$$

$$i_{(R_1)} = \frac{V}{R} = \frac{120 - 50}{5 \times 10^3} = \frac{70}{5 \times 10^3} = 14 \times 10^{-3} \text{ A}$$

$$\therefore i_{\text{zener diode}} = 14 \times 10^{-3} - 5 \times 10^{-3} = 9 \times 10^{-3} \text{ A} = 9 \text{ mA}$$

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Q5

(1) Since voltage across zener diode does not reach to breakdown voltage therefore its resistance will be infinite & current through it is 0.

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Q6

(2) As D_2 is reversed biased, so no current through 75Ω resistor.

$$\text{now } R_{eq} = 150 + 50 + 100 \\ = 300 \Omega$$

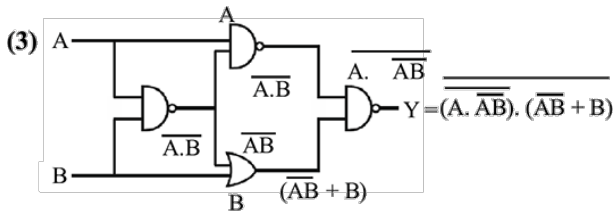
$$\text{So, required current } I = \frac{\text{Battery Voltage}}{300}$$

$$I = \frac{6}{300} = 0.02$$

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Q7

Semiconductors



$$Y = (\overline{A \cdot \overline{AB}}) + (\overline{AB + B})$$

$$= A \cdot \overline{AB} + AB \cdot \overline{B}$$

$$= A \cdot (\overline{A} + \overline{B}) + AB \cdot \overline{B}$$

$$= A\overline{B}$$

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Q8

(1) At saturation, $V_{CE} = 0$

$$V_{CE} = V_{CC} - I_C R_C$$

$$\Rightarrow I_C = \frac{V_{CC}}{R_C} = 5 \times 10^{-3} \text{ A}$$

Current gain,

$$\beta_{dc} = \frac{I_C}{I_B}$$

$$I_B = \frac{5 \times 10^{-3}}{200} = 25 \mu\text{A}$$

At input side

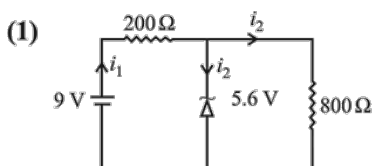
$$V_{BB} = I_B R_B + V_{BE}$$

$$= (25 \text{ mA})(100 \text{ k}\Omega) + 1 \text{ V}$$

$$V_{BB} = 3.5 \text{ V}$$

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Q9



P.D. across 800Ω resistors = 5.6 V

$$\text{so, } I_{800\Omega} = \frac{5.6}{800} \text{ A} = 7 \text{ mA}$$

Now, P.D. across 200Ω resistors = $(9 - 5.6) \text{ V} = 3.4 \text{ V}$

$$\text{so, } I_{200\Omega} = \frac{9 - 5.6}{200} = 17 \text{ mA}$$

$$\text{so, current through zener diode}$$

$$= I_2 = 17 - 7 = 10 \text{ mA}$$

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Q10

(1) Given, $\beta = 250$

$$\text{Voltage gain, } \frac{V_{CC}}{V_B} = \beta \frac{R_C}{R_B}$$

$$\frac{10}{V_B} = 250 \times \frac{10^3}{R_B}$$

$$\therefore \frac{V_B}{R_B} = \frac{1}{25 \times 10^3} = 40 \mu\text{A}$$

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Q11

(2) Given $\Delta V_i = 10 \times 10^{-3} \text{ V}$

$$\Delta I_c = 3 \times 10^{-3} \text{ A}$$

$$\Delta I_b = 15 \times 10^{-6} \text{ A}$$

$$R_i = \frac{\Delta V_i}{\Delta I_b} = \frac{10 \times 10^{-3}}{15 \times 10^{-6}} = 0.67 \text{ k}\Omega$$

$$\therefore \text{Voltage gain} = \frac{\Delta I_c}{\Delta I_b} \times \frac{R_0}{R_i}$$

$$= \left(\frac{3 \times 10^{-3}}{15 \times 10^{-6}} \right) \times \frac{1000}{670} = 200 \times \frac{1000}{670} \approx 300$$

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Q11

(2) Truth table $\rightarrow$

A	B	$\overline{A}$	$\overline{B}$	$\overline{A \cdot B}$
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

The output is of OR-gate

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Q12

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(1) Power gain = 60 = $10 \log \left(\frac{P_0}{P_i} \right)$

$$\Rightarrow 6 = \log \left(\frac{P_0}{P_i} \right)$$

$$\Rightarrow \frac{P_0}{P_i} = 10^6$$

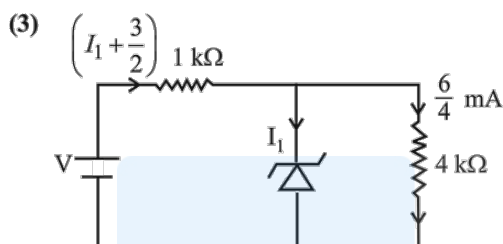
$$= \beta^2 \left(\frac{R_{out}}{R_{in}} \right)$$

$$\Rightarrow 10^6 = \beta^2 \left(\frac{10000}{100} \right) \quad \left[\begin{array}{l} \text{as } R_{out} = 10,000 \Omega \\ R_{in} = 100 \Omega \end{array} \right]$$

$$\Rightarrow \beta = 100$$

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Q13



For voltage, $V = 8 \text{ V}$

$$\text{Current, } I_1 = \left(8 - 6 - \frac{3}{2} \right) = \frac{1}{2} = 0.5 \text{ mA}$$

For voltage, $V = 16 \text{ V}$

$$\text{Current, } I_2 = \left(16 - 6 - \frac{3}{2} \right) = 8.5 \text{ mA}$$

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Q14

(3)

A	B	(A + B)	(A + B). A	$\overline{(A + B). A}$
0	0	0	0	1
0	1	1	0	1
1	0	1	1	0
1	1	1	1	0

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Q14

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(Bonus) $\beta = \frac{\Delta i_c}{\Delta i_b} = \frac{200 - 100}{10 - 5} = 20$

$$\text{Voltage gain} = \beta \frac{R_2}{R_1} = \frac{20 \times 100 \times 10^3}{100} = 20 \times 10^3$$

$$\text{Power gain} = \beta^2 \frac{R_2}{R_1} = 20^2 \left(\frac{100 \times 10^3}{100} \right) = 4 \times 10^5$$

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Q15

(4) Current in load resistance,

$$i_1 = \frac{6}{4 \times 10^3} = 1.5 \times 10^{-3} \text{ A} = 1.5 \text{ mA}$$

For $V = 16 \text{ volt}$,

$$i_s = \frac{(16 - 6)}{2 \times 10^3} = 5 \text{ mA}$$

$$\therefore i_2 = i_s - i_1 = 5 - 1.5 = 3.5 \text{ mA}$$

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